

Directing the Generative Cut

Cusuh Ham

Gen4AVC Workshop

September 8, 2026

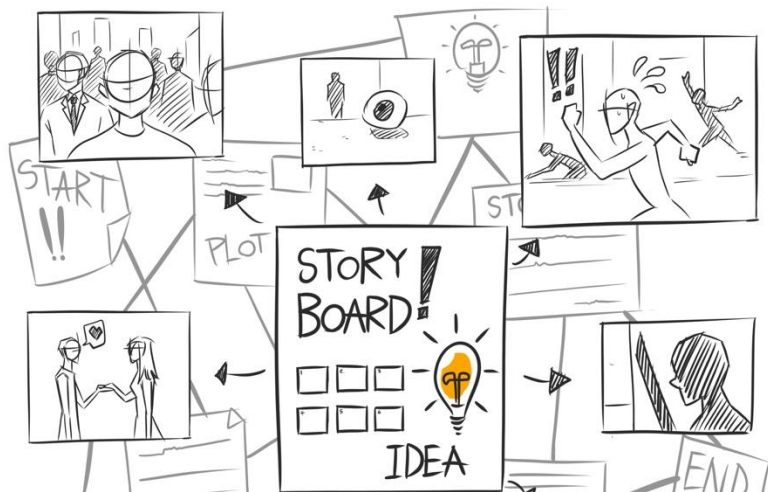


Creators often have a specific vision in mind

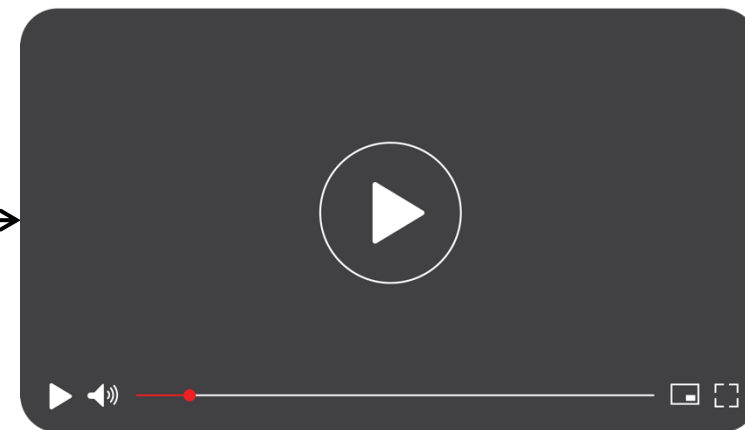


From intent to specification

Creative intent



Generated video



- Motion
- Timing
- Framing
- Composition
- Subject
- Setting
- Style
- ...

Shot specification

Directing the motion



Directing the motion



Grounding the cut



Sequence 01

00:01:03:05

00:00:59:22 00:01:04:22 00:01:09:22

Vince exiting a wooden cabin

Veo 3.1 1080p 16:9 24fps 8 sec

First frame 0/3 Uses 400 credits Generate

06:23

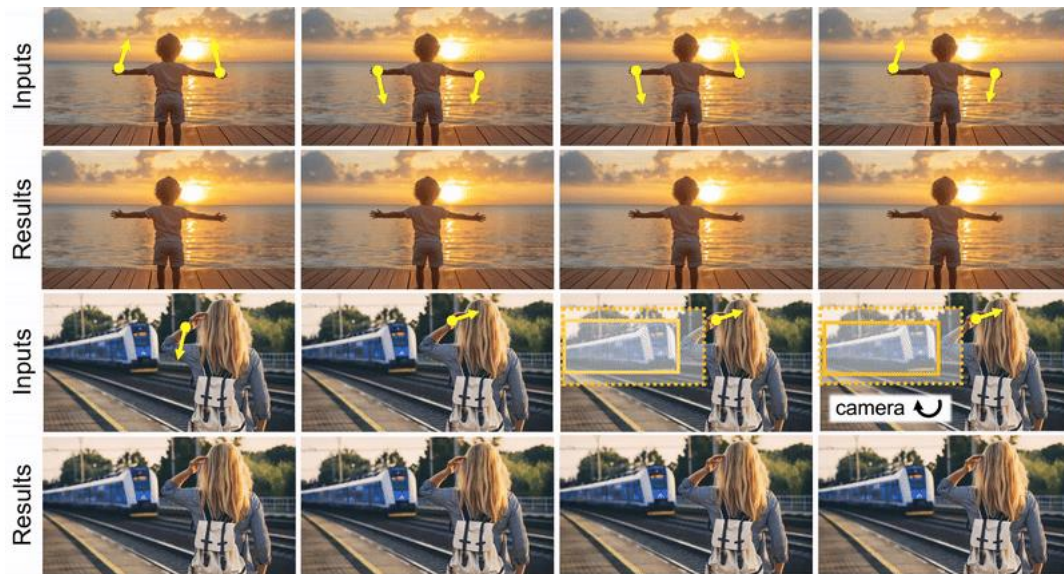
VincesHome_Surro... Vinc...

VinceMcIntyre_H... VinceMcInt... VinceMcInt... Vi...

V1 V2 V3

A1 A2 A3 Mix

0.0



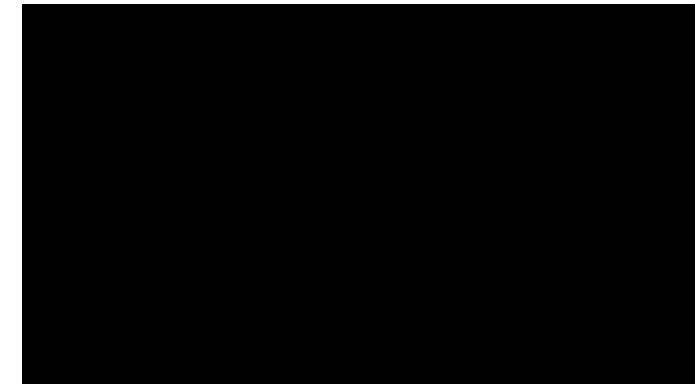
MotionCanvas: Cinematic Shot Design with
Controllable Image-to-Video Generation
SIGGRAPH 2025

Video collection



User request

*"Sequence showing the
man's daily activities
[...]" (24s)*



Memory-Guided B-Roll Generation from
User Video Collections
SIGGRAPH Asia 2026

MotionCanvas: Cinematic Shot Design with Controllable Image-to-Video Generation

Jinbo Xing, Long Mai, Cusuh Ham, Jiahui Huang, Aniruddha Mahapatra, Chi-Wing Fu,
Tien-Tsin Wong, Feng Liu

SIGGRAPH 2025

Adobe

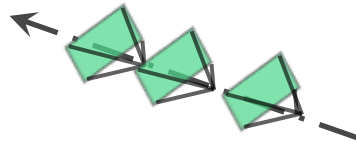


Shot design





2D *screen* space



[Camera: dolly-in]

3D *scene* space





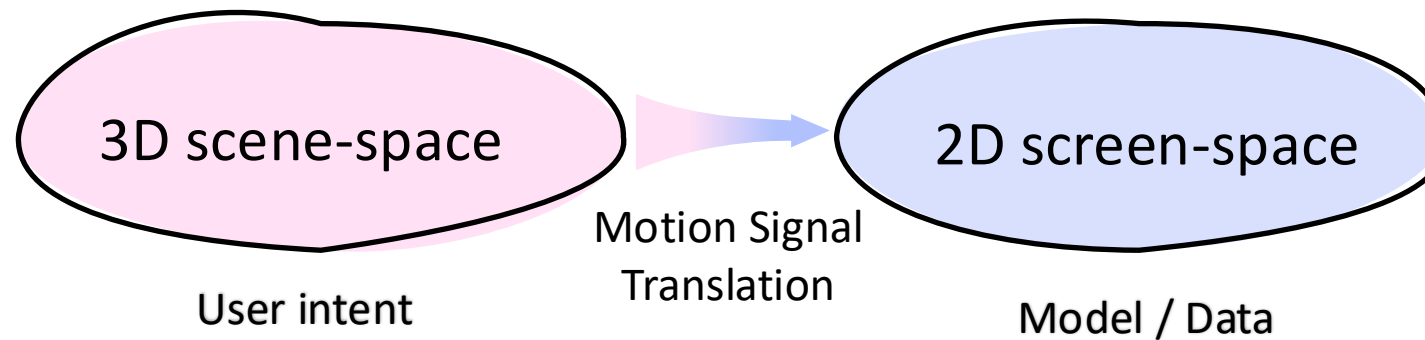
3D vs 2D?

3D

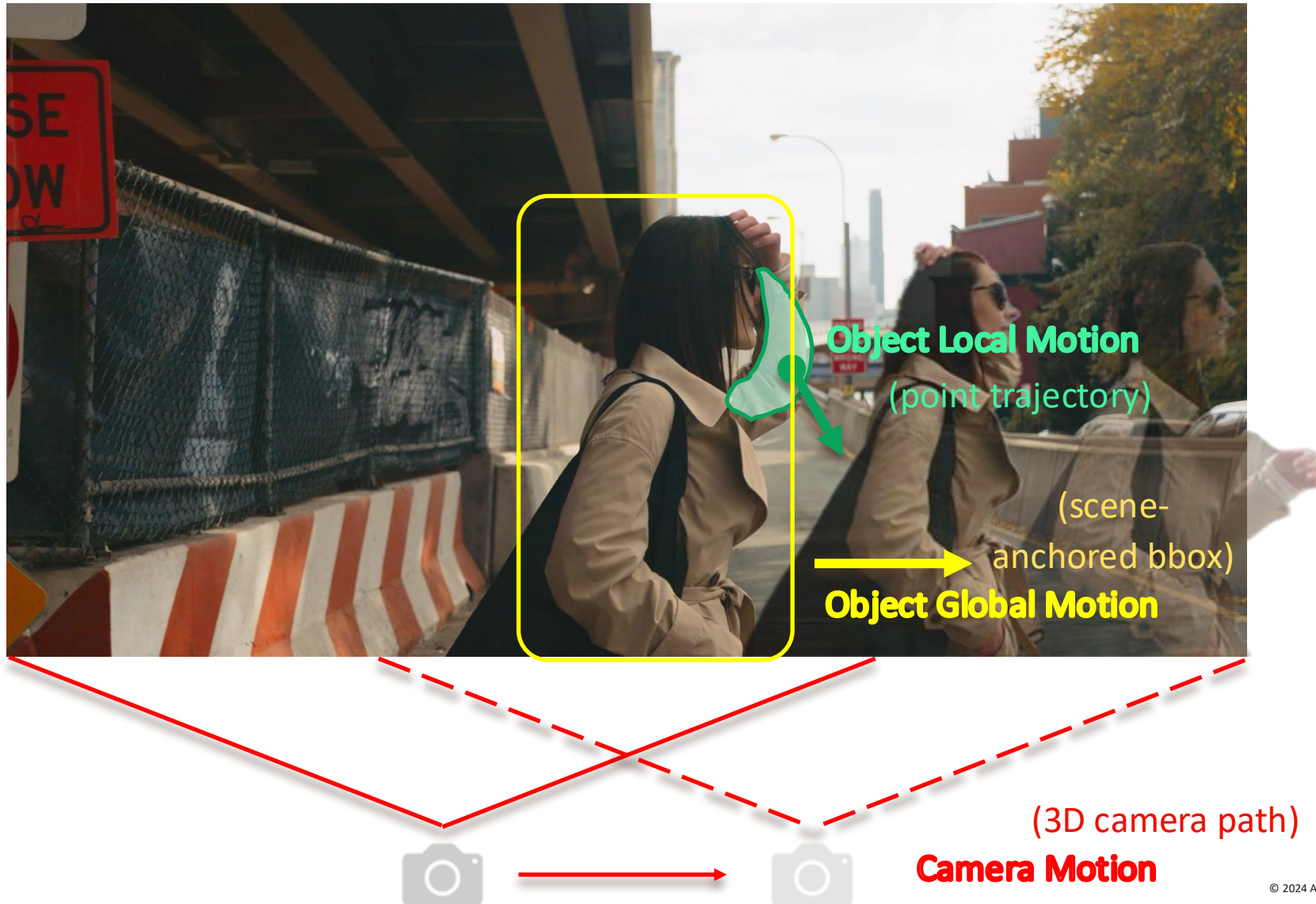
- + Intuitive for users
- Unreliable labeling
- Limited data

2D

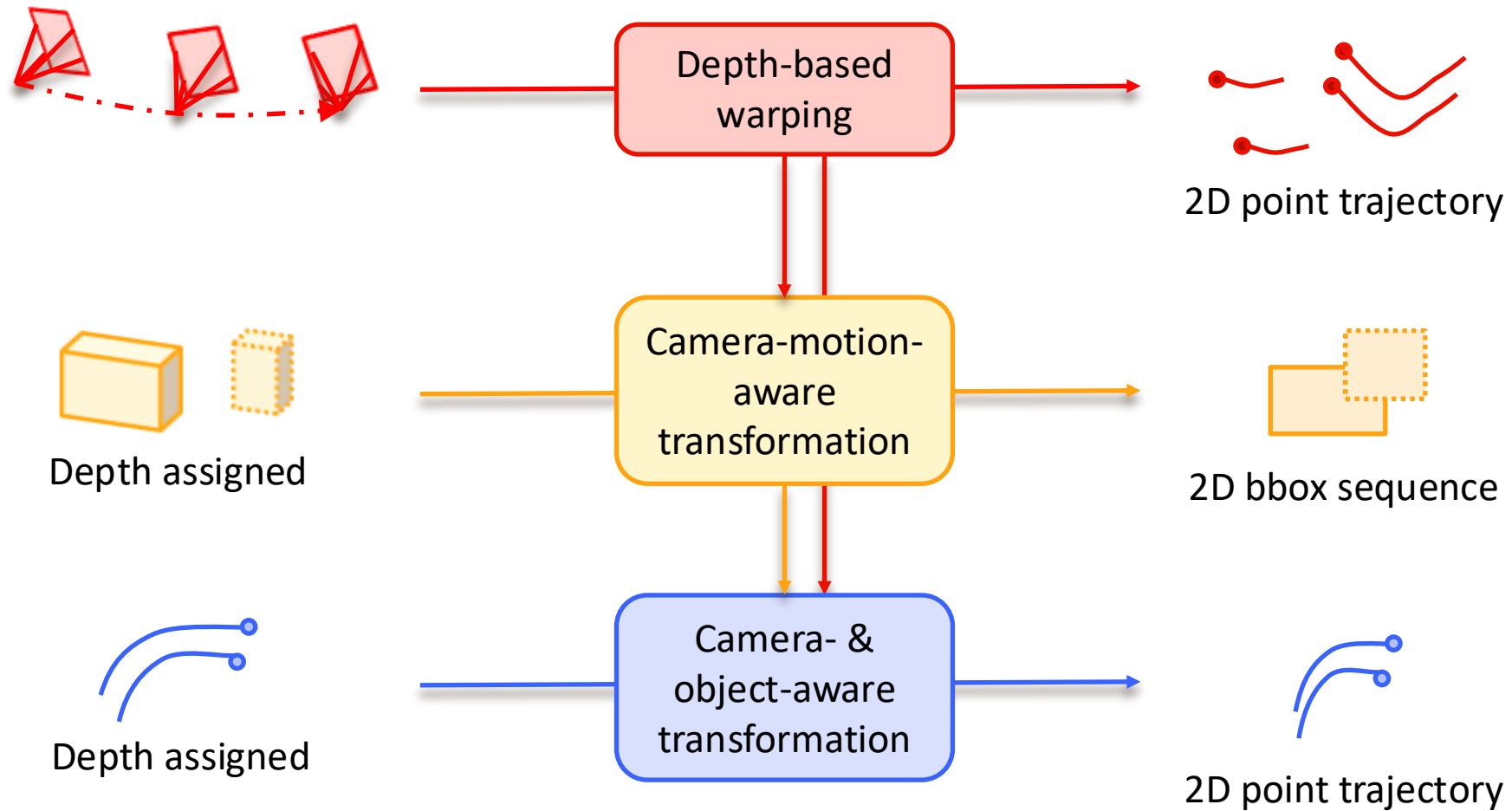
- Unintuitive for users
- + Reliable labeling
- + Scalable data
- + Effective for training video models



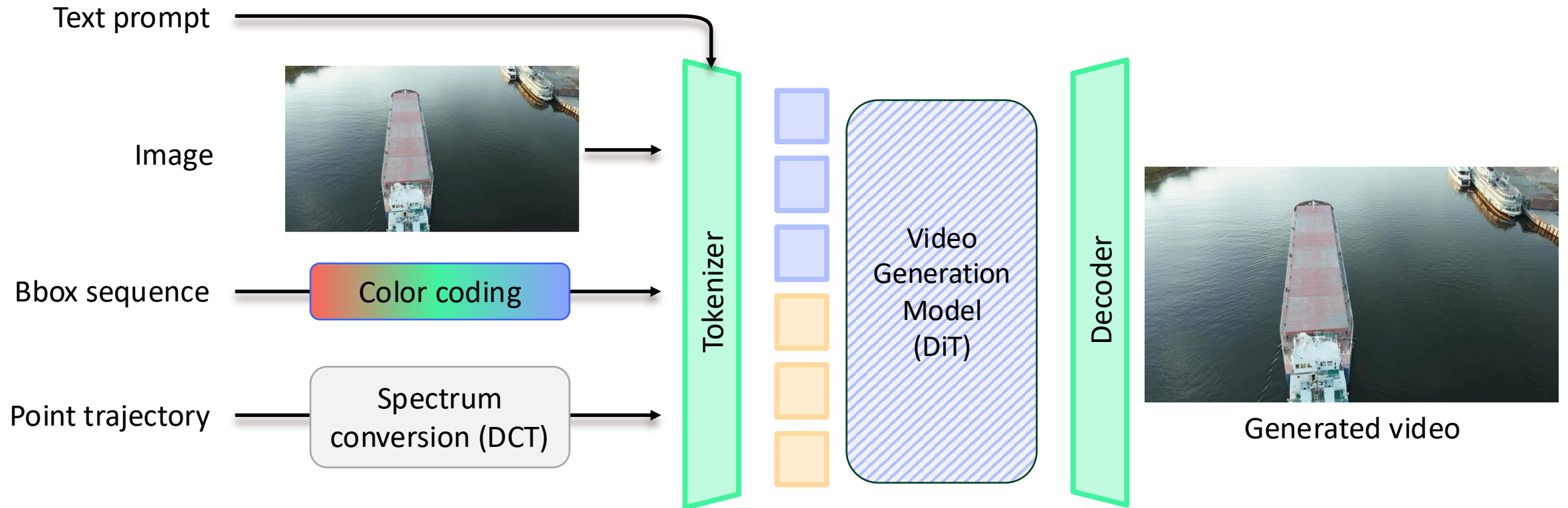
Shot design



Motion signal translation



MotionCanvas training



Users can flexibly control both camera motion and object motion

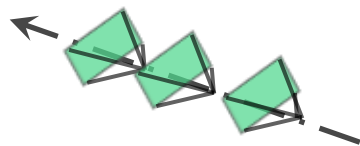
Row: different camera motions with the same object motion

Column: same camera motion with different object motions

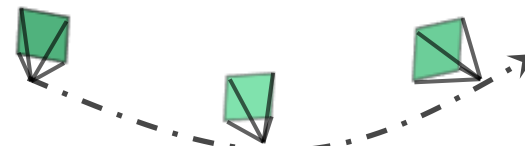
Object Motion



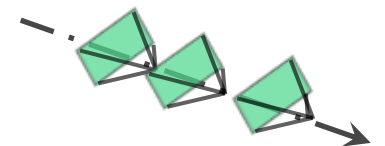
Keep selected objects static



[Camera: dolly-in]



[Camera: orbit-right]



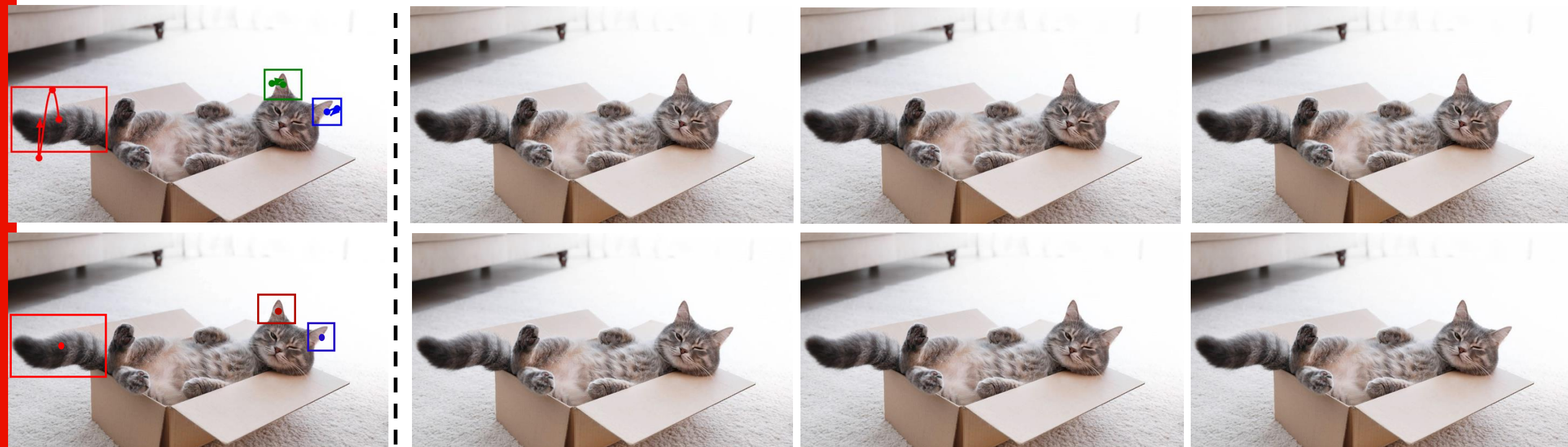
[Camera: dolly-out]

Users can flexibly control both camera motion and object motion

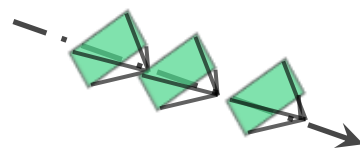
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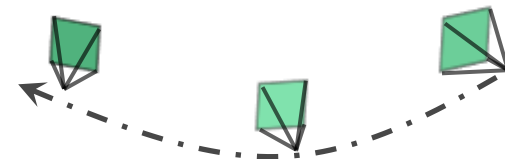
Object Motion



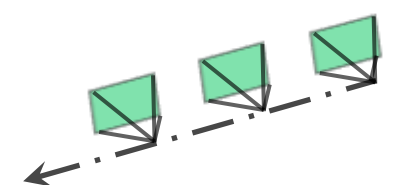
Keep selected objects static



[Camera: dolly-out]

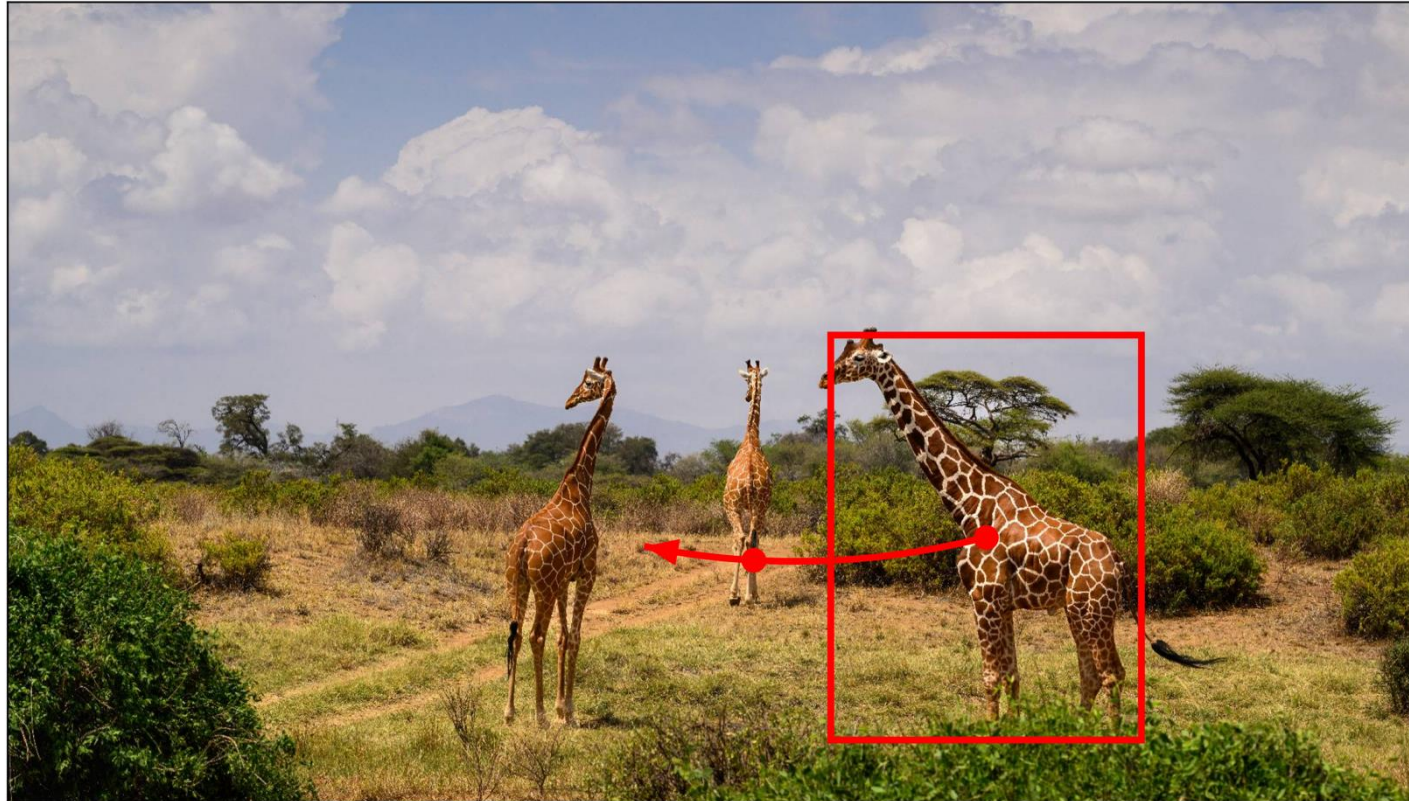


[Camera: orbit-left]



[Camera: diagonal-left]

Results





Memory-Guided B-Roll Generation from User Video Collections

Cusuh Ham, Fabian Caba Heilbron, Josef Sivic, Bryan Russell

SIGGRAPH Asia 2026



Missing shot

User video collection

Sequence 01

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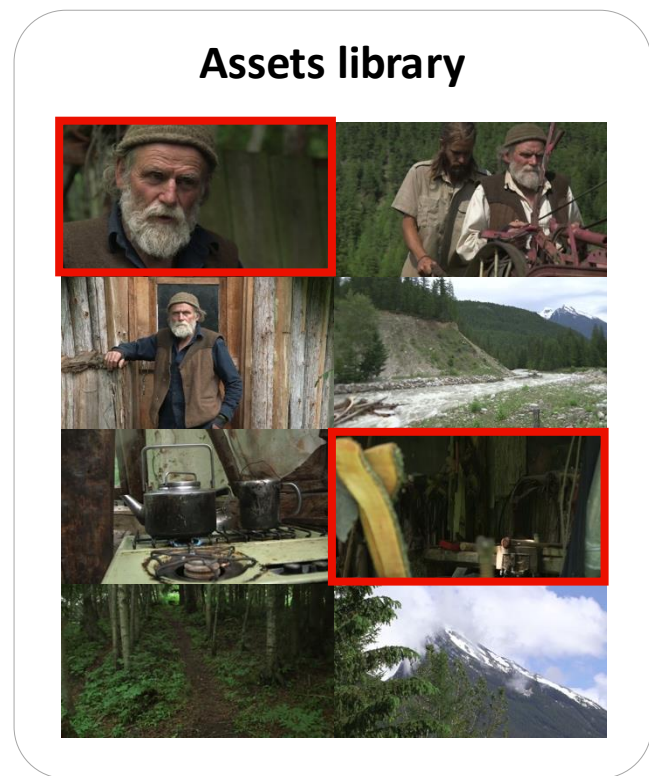
VinceMcIntyre_H... VinceMcInt... VinceMcInt... VinceMcInt... Vi...

Vince...

A1 A2 A3 Mix

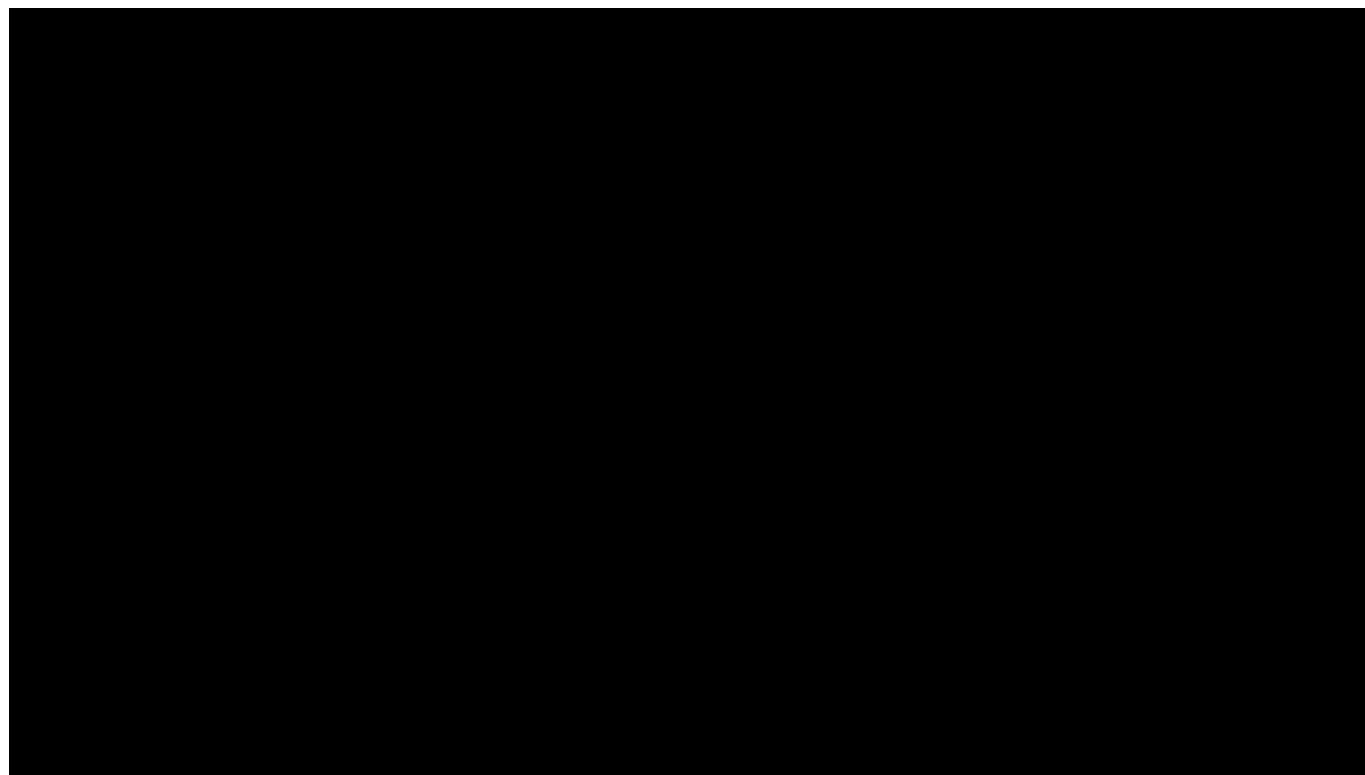
1 1 1 1 1 1 1 1 1

Generative models are powerful but require good references



Video
generation
model

Output generated clip



"Close-up of a man's weathered hands using a drawknife on a shaving horse to peel bark from a branch inside a rustic workshop."

 = selected reference frames

Manual reference selection is not scalable

Assets library

A few frames



User video collection

499 clips, ~10.5 hrs of captured footage



Key insight: condensed memory representation

User video collection

499 clips, ~10.5 hrs of captured footage



Memory

(~8% frames)

Collection summary

"This collection follows an elderly homesteader and his associates living a self-sufficient lifestyle [...]."

Frames

Characters



Settings



Character sheet



char_00: Caucasian male, late 60s/early 70s;
long, thick, bushy white beard



char_01: Caucasian male, late 20s/early 30s;
long, sandy-blond hair

MemComposer

Output generated B-roll sequence

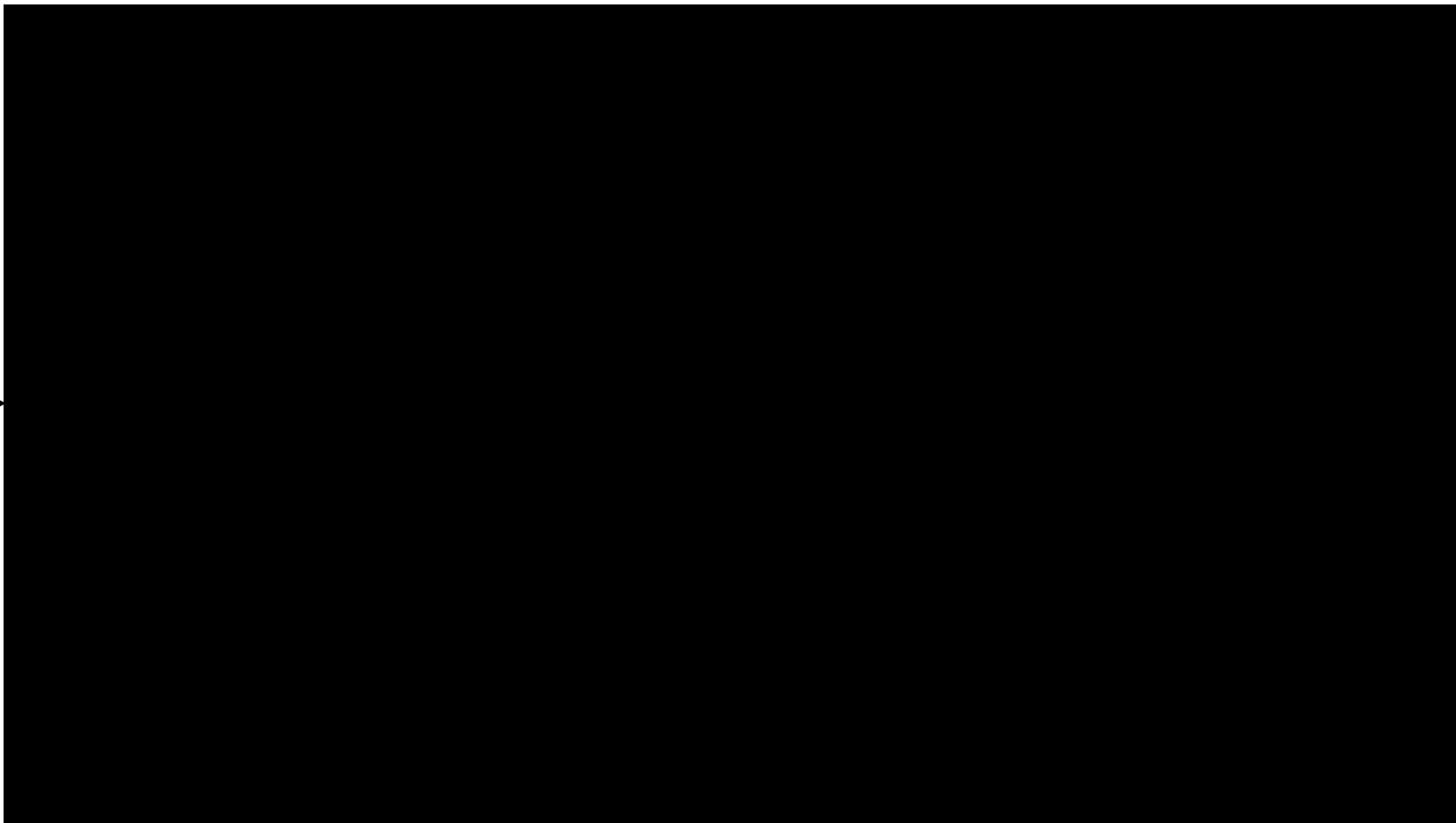
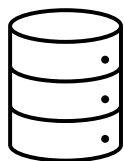
Inputs

User request

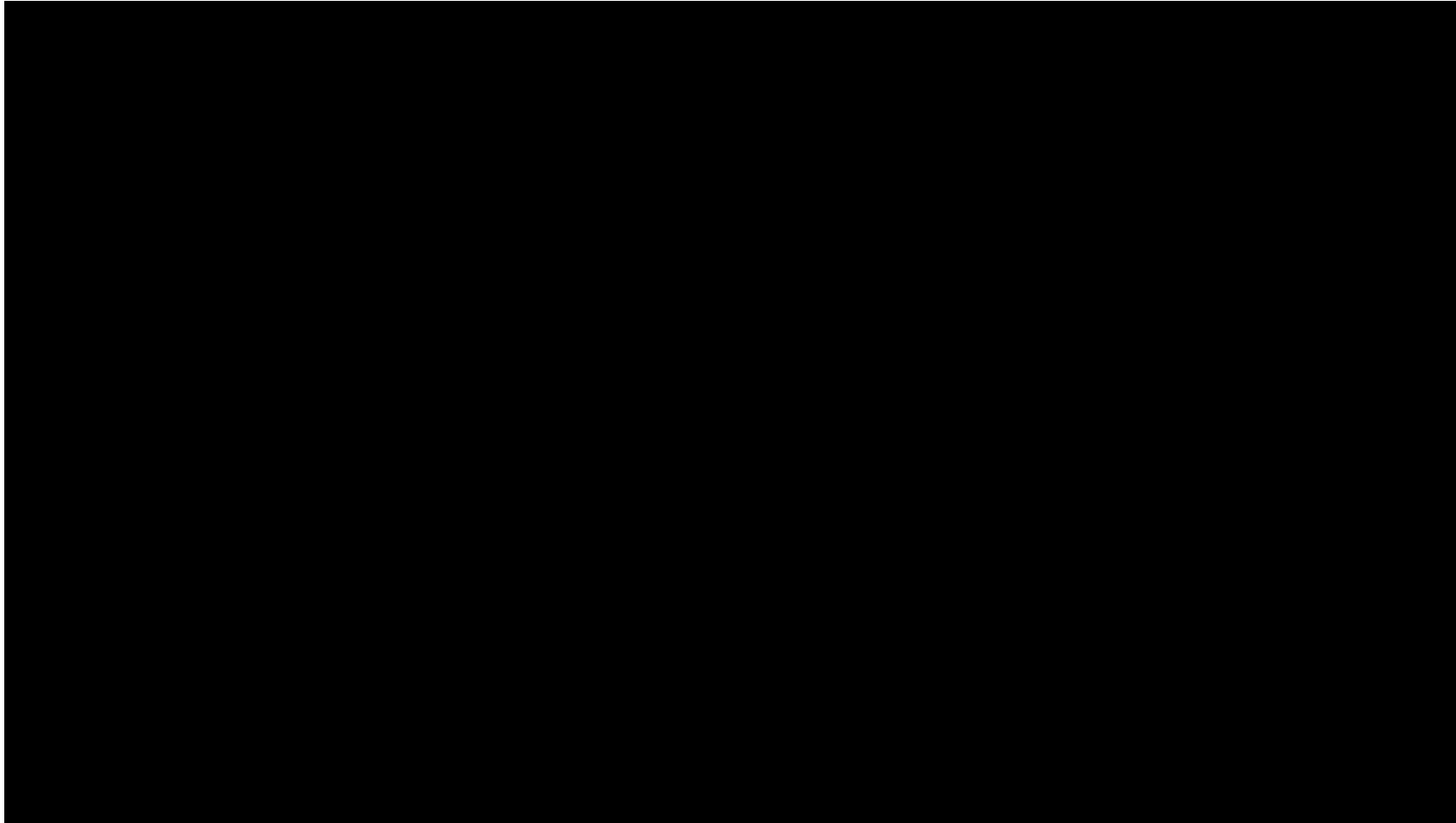
"Sequence showing the man's daily activities. Start from him walking in the woods, and quickly transition to his arrival into a workshop. Show a mixed of interleaved close-ups of decoration, his hands, and his face."

(24s)

Memory (~8% frames)



Synthesizing missing footage

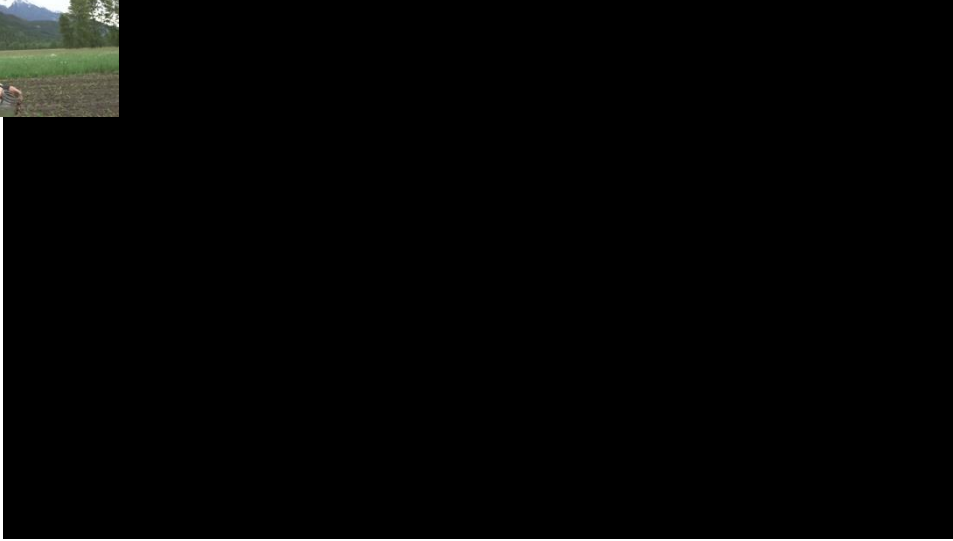


*"Sequence showing **Vince**'s daily activities alongside **the young man**. Interleave close-ups of them working, including a moment where they exchange brief eye contact." (24s)*

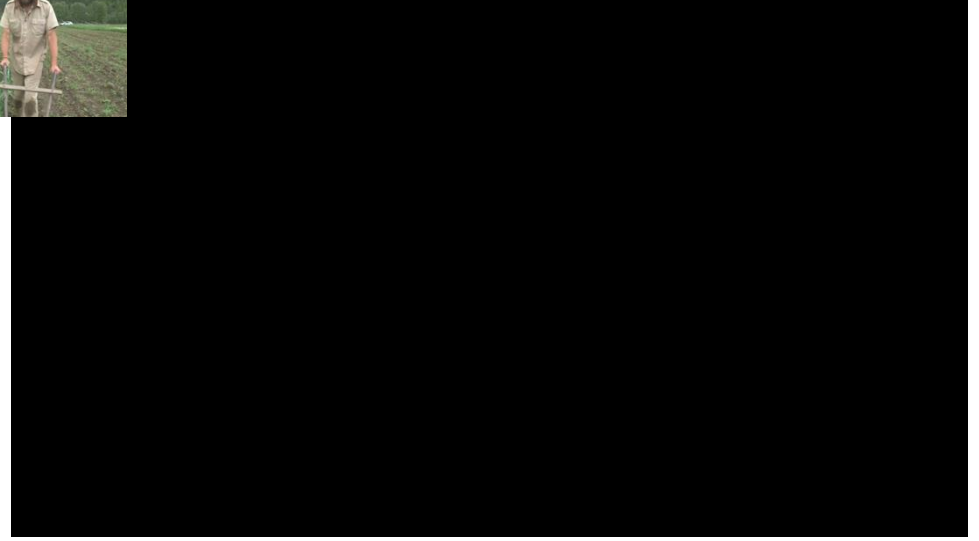
Swapping characters across different collections



Built By Life



Built By Life



The Scramble King



The Rock Climber

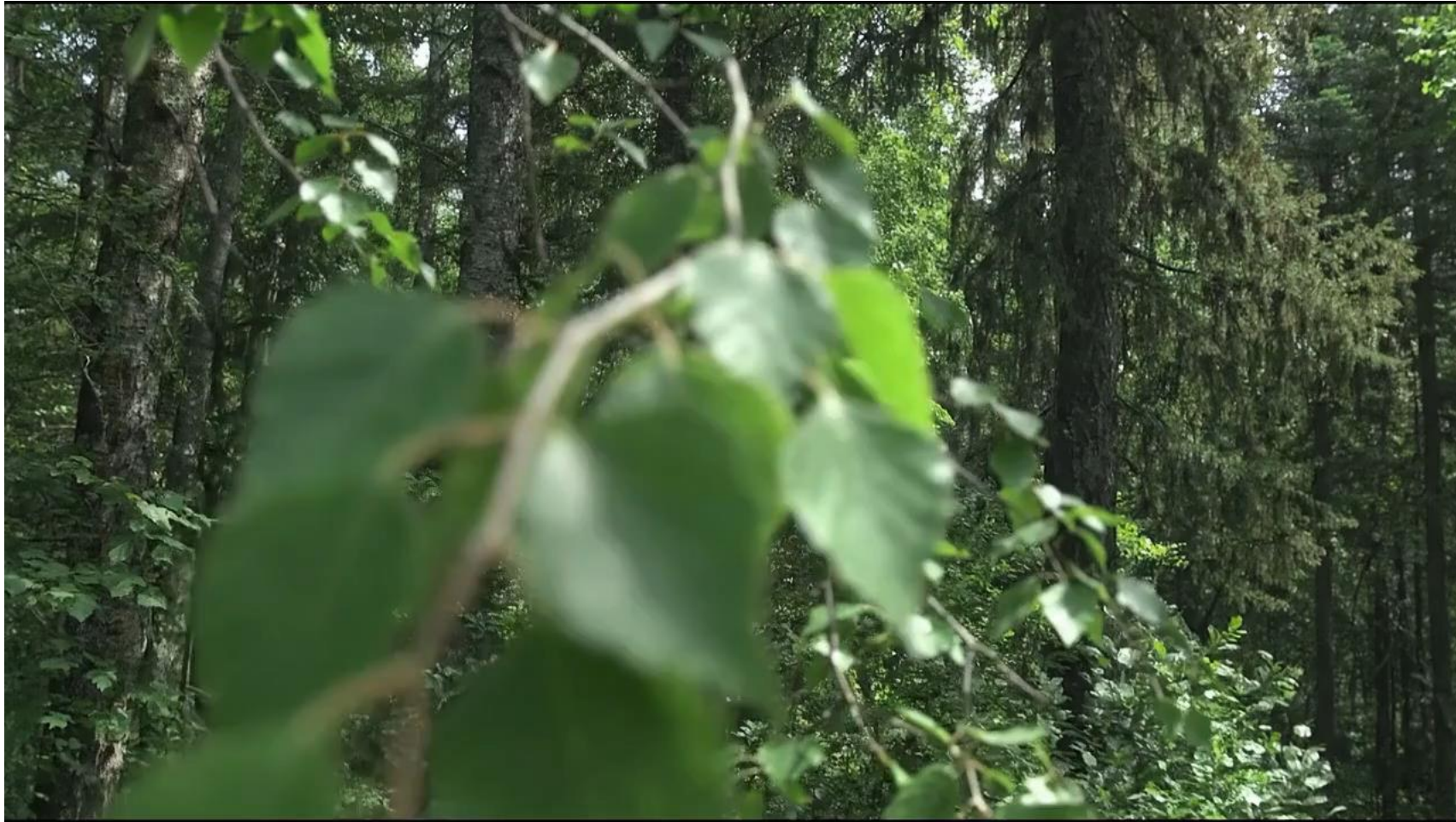


Context is not only visual



*"Montage of nature ending at **Vince**'s wooden house. Start with an out of focus shot of leaves on a tree that slowly comes into focus. End with a shot of the man leaving his home." (24s)*

Context is not only visual



*"Montage of nature ending at **Vince**'s wooden house. Start with an out of focus shot of leaves on a tree that slowly comes into focus. End with a shot of the man leaving his home." (24s)*

Generation should not just make plausible clips; it should help creators produce the shot they envision.



Thank you! Questions?